Quantum Mechanics Problem Sheet Solutions

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Abstract

This paper presents solutions to the quantum mechanics problem sheets provided alongside the 2026 British Physics Olympiad Computational Physics Challenge. Derivations, physical interpretations and relevant computational visualisations are included where appropriate.

Contents

Quantum Mechanics 1

Question 1

(i)

Problem

To an order of magnitude (that is, to the nearest power of ten), what is the radius of an atom in metres? How does this compare with the radius of an atomic nucleus?

Solution

A typical atomic radius is $r_{\text{atom}} \sim 10^{-10}\,\text{m}$, whereas a nuclear radius is $r_{\text{nuc}} \sim 10^{-15}\,\text{m}$. Hence

$$\frac{r_{\text{atom}}}{r_{\text{nuc}}} \sim 10^5, \qquad \frac{V_{\text{atom}}}{V_{\text{nuc}}} \sim 10^{15}.$$

Thus almost all atomic volume is occupied by the electron cloud rather than nuclear matter.

Computational link: Task 10 turns this scale separation into probability-density plots: the nucleus is effectively point-like on the Ångström-scale grids used for hydrogenic orbitals.

(ii)

Problem

A large blue marble has diameter about 3.6 cm. Contrast the number of atoms that constitute the marble with the number of such marbles that could fill an Earth-sized container. Take the Earth's diameter to be 1.28×10^7 m.

Solution

Using a characteristic atomic diameter $d_a \sim 10^{-10}\,\text{m}$,

$$N_a \sim \left(\frac{3.6 \times 10^{-2}}{10^{-10}}\right)^3 = 4.7 \times 10^{25}.$$

For Earth diameter $1.28 \times 10^7\,\text{m}$,

$$N_m \sim \left(\frac{1.28 \times 10^{7}}{3.6 \times 10^{-2}}\right)^3 = 4.5 \times 10^{25}.$$

Therefore $\boxed{N_a \sim N_m \sim 5 \times 10^{25}}$

Packing and material-density corrections alter only the prefactor.

Question 2

Problem

Describe briefly the Rutherford (Geiger–Marsden) experiment, including a suitable diagram, and state what its observations imply about atomic structure.

Solution

A collimated beam of α-particles was directed at thin gold foil; a zinc-sulfide screen recorded scattering angles.

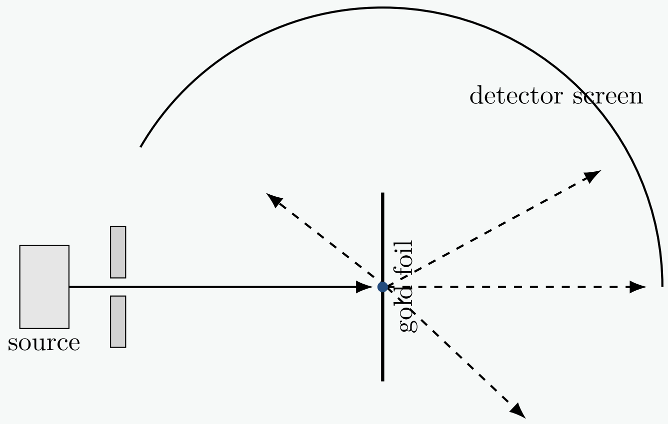

Figure 1: Geiger–Marsden scattering geometry.

Most particles passed straight through, some were moderately deflected, and a tiny fraction were backscattered. Therefore most of the atom is empty space, while nearly all positive charge and mass occupy a tiny nucleus. Close approaches give strong Coulomb repulsion,

$$F = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r^2}.$$

The experiment established the nuclear atom, not the later quantum description of orbitals.

Question 3

(i)

Problem

An old 100 W light bulb produces light using a glowing tungsten filament, modelled as a thin strip of length 20 cm and width 0.14 mm. Calculate its temperature in degrees Celsius. State the assumptions made and comment on whether they are reasonable.

Solution

Following the sheet's effective radiating-area convention, $A = lw = 2.8 \times 10^{-5}\,\mathrm{m}^2$. Stefan's law gives

$$T = \left(\frac{P}{\sigma A}\right)^{1/4} = 2.82 \times 10^3\,\mathrm{K} = \boxed{2.54 \times 10^3\,{}^\circ\mathrm{C}}$$

Assumptions: the full electrical power is radiated, emissivity is one, temperature is uniform, and conductive/convective losses are negligible. If both broad faces are counted separately, the estimate becomes 2.37×10^3 K; this exposes the ambiguity in the simple strip model.

Computational link: Task 3 evaluates the full Planck spectrum; integrating that spectrum over wavelength recovers the T^4 scaling used here.

(ii)

Problem

The Sun has radius 695 500 km and surface temperature approximately 5778 K. Calculate its luminosity in watts.

Solution

$$L_\odot = 4\pi R_\odot^2 \sigma T_\odot^4 = \boxed{3.84 \times 10^{26}\,\mathrm{W}}$$

(iii)

Problem

Calculate the width of a square solar panel that delivers a maximum electrical power of 100 W on the surface of Mars. Mars is 227.9 million kilometres from the Sun and the panel is 20% efficient.

Solution

$$I_M = \frac{L_\odot}{4\pi r^2} = 5.89 \times 10^2\,\mathrm{W\,m^{-2}}, \quad A = \frac{100}{0.20 I_M} = 0.849\,\mathrm{m^2},$$

so $\boxed{s = \sqrt{A} = 0.922\,\mathrm{m}}$, for normal incidence and no extra losses.

(iv)

Problem

Human core temperature is approximately 37°C, while surface skin temperature is about 34°C. Determine λ_{max}, in nanometres, for radiation emitted by human skin and identify the relevant part of the electromagnetic spectrum.

Solution

With $T = 307\,\mathrm{K}$, Wien's law gives

$$\lambda_{\mathrm{max}} = \frac{2.90 \times 10^{-3}}{307} = \boxed{9.45 \times 10^{-6}\,\mathrm{m} = 9.45\,\mu m},$$

in the thermal infrared.

(v)

Problem

Sketch the radiation spectrum of the Sun together with those of a cooler red star and a hotter blue star. Calculate λ_{max} for the Sun in nanometres.

Solution

$$\lambda_{\mathrm{max},\odot} = \frac{2.90 \times 10^{-3}}{5778} = \boxed{5.02 \times 10^{-7}\,\mathrm{m} = 502\,\mathrm{nm}}$$

The hotter spectrum peaks at shorter wavelength and greater radiance; the cooler spectrum peaks at longer wavelength.

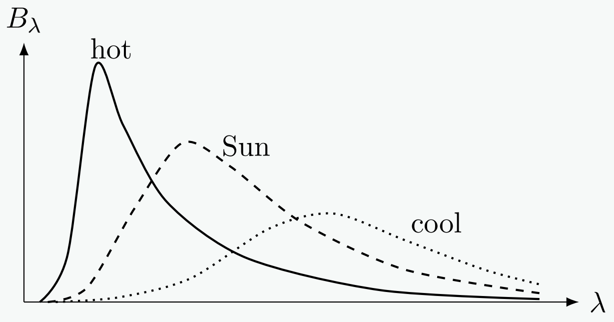

Figure 2: Qualitative black-body spectra.

Computational link: Task 3 plots $B_\lambda(\lambda, T)$ for several temperatures, so the movement and growth of the peak can be checked directly against Wien's law.

(vi)

Problem

At the beginning of the Sun's expansion into a red giant, assume that its luminosity is unchanged. Calculate the ratio of the new solar radius to its present radius if $\lambda_{\max} = 680\,\mathrm{nm}$.

Solution

Wien's law gives $T_{\mathrm{RG}} = 4.26 \times 10^3\,\mathrm{K}$. Since $L \propto R^2 T^4$,

$$\frac{R_{\mathrm{RG}}}{R_\odot} = \left(\frac{T_\odot}{T_{\mathrm{RG}}}\right)^2 = \boxed{1.84}$$

This is an early-stage, fixed-luminosity idealisation.

Question 4

(i)

Problem

Silver has work function $4.3\,\mathrm{eV}$. Calculate the maximum wavelength, in nanometres, that can produce photoelectrons.

Solution

At threshold the maximum kinetic energy is zero, so Einstein's equation gives

$$\phi = hf_0 = \frac{hc}{\lambda_0}.$$

Converting the work function,

$$\phi = 4.3(1.60 \times 10^{-19}) = 6.88 \times 10^{-19}\,\mathrm{J}.$$

Hence

$$\lambda_0 = \frac{hc}{\phi} = \boxed{2.89 \times 10^{-7}\,\mathrm{m} = 289\,\mathrm{nm}},$$

which lies in the ultraviolet.

Computational link: Task 4 uses each metal's work function to set its threshold frequency and wavelength, below which the plotted stopping voltage is clamped to zero.

(ii)

Problem

Ultraviolet light of wavelength $200\,\mathrm{nm}$ is incident on silver. Calculate the maximum kinetic energy of the emitted photoelectrons in joules.

Solution

The photon energy is

$$E_\gamma = \frac{hc}{\lambda} = 9.94 \times 10^{-19}\,\mathrm{J} = 6.21\,\mathrm{eV}.$$

Einstein's photoelectric equation then gives

$$K_{\max} = E_\gamma - \phi = (6.21 - 4.30)\,\mathrm{eV} = \boxed{3.06 \times 10^{-19}\,\mathrm{J}}$$

This is the maximum kinetic energy; electrons emitted from below the surface may emerge with less.

(iii)

Problem

The ultraviolet power absorbed by the silver is $1.00\,\mu$W. Assuming all incident light is absorbed, calculate the number of photons absorbed each second.

Solution

Power is energy transferred per unit time. Since each photon has energy $E_\gamma = hc/\lambda$,

$$P = \dot{N}_\gamma E_\gamma \quad \Rightarrow \quad \dot{N}_\gamma = \frac{P}{hc/\lambda}.$$

Using $P = 1.00 \times 10^{-6}\,\mathrm{W}$ and the photon energy from part (ii),

$$\boxed{\dot{N}_\gamma = 1.01 \times 10^{12}\,\mathrm{s}^{-1}}$$

Computational link: Task 4's intensity control changes the photon arrival rate and hence the saturation current, without changing the stopping voltage.

(iv)

Problem

Assuming that one photoelectron is produced per absorbed photon, calculate the photocurrent. Use $e = 1.60 \times 10^{-19}\,\mathrm{C}$.

Solution

One emitted electron per photon gives $\dot{N}_e = \dot{N}_\gamma$. Therefore

$$I = \frac{\Delta Q}{\Delta t} = e\dot{N}_e = (1.60 \times 10^{-19})(1.01 \times 10^{12}),$$

so

$$\boxed{I = 1.61 \times 10^{-7}\,\mathrm{A} = 0.161\,\mu A}$$

This is an ideal maximum because it assumes unit quantum efficiency and complete charge collection.

(v)

Problem

Calculate the stopping voltage required to reduce the photocurrent to zero.

Solution

At the stopping potential, the electric potential-energy change equals the largest electron kinetic energy:

$$eV_s = K_{\max}.$$

Hence

$$V_s = \frac{3.06 \times 10^{-19}\,\mathrm{J}}{1.60 \times 10^{-19}\,\mathrm{C}} = \boxed{1.91\,\mathrm{V}}$$

The same numerical value follows because $K_{\max} = 1.91\,\mathrm{eV}$.

Computational link: Task 4 implements $V_s = (hf - \phi)/e$; wavelength sets the stopping voltage while intensity sets the available current.

Question 5

(i)

Problem

An electron of mass $m = 9.109 \times 10^{-31}\,\mathrm{kg}$ and charge magnitude e is accelerated from rest through a potential difference V. Find its final speed in terms of m, e and V.

Solution

The electrical work done on the electron is eV. Starting from rest and using classical kinetic energy,

$$eV = \frac{1}{2}mv^2.$$

Therefore

$$v^2 = \frac{2eV}{m}, \qquad \boxed{v = \sqrt{\frac{2eV}{m}}}$$

The positive root is used because v denotes speed.

Computational link: Task 6 begins with this voltage-to-speed conversion before replacing speed by momentum and de Broglie wavelength.

(ii)

Problem

Derive an inequality for V such that the electron speed is less than 1% of $c = 2.998 \times 10^8\,\mathrm{m\,s^{-1}}$. State what physical correction can then be ignored.

Solution

The requirement is

$$\sqrt{\frac{2eV}{m}} < 0.01c.$$

Squaring and rearranging gives

$$V < \frac{10^{-4}mc^2}{2e} = \boxed{25.6\,\mathrm{V}}$$

At such speeds $\gamma - 1 \lesssim 5 \times 10^{-5}$, so relativistic corrections to $p = mv$ and $K = mv^2/2$ are negligible. Quantum wave behaviour is not thereby negligible.

(iii)

Problem

In the classical limit, show that the electron's de Broglie wavelength is

$$\lambda = \frac{h}{\sqrt{2meV}}.$$

Solution

From part (i),

$$p = mv = m\sqrt{\frac{2eV}{m}} = \sqrt{2meV}.$$

Substitution into the de Broglie relation $\lambda = h/p$ gives

$$\boxed{\lambda = \frac{h}{\sqrt{2meV}}}$$

Thus $\lambda \propto V^{-1/2}$, the dependence used in the electron-diffraction model.

Computational link: Task 6 uses this $V^{-1/2}$ dependence to update the diffraction angles and ring radii as the accelerating-voltage slider moves.

(iv)

Problem

An electron beam strikes a carbon disc whose atomic layers are approximately 1.23×10^{-10} m apart. Sketch electron intensity on a screen against angle from the original beam direction. Explain the pattern and what it reveals about electrons. Consider Bragg scattering from alternate atomic layers.

Solution

Constructive interference occurs when

$$d \sin\phi = n\lambda,$$

using the effective spacing convention of the sheet. Random crystal orientations generate diffraction cones, seen as rings on a planar screen. Localised impacts build a wave-like interference pattern.

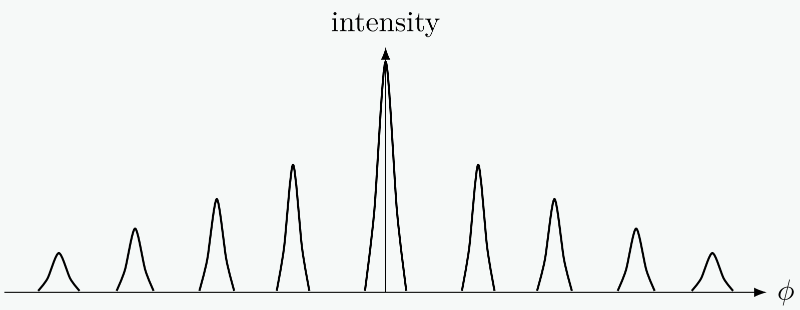

Figure 3: Qualitative symmetric diffraction peaks.

Computational link: Task 6 converts these diffraction cones into concentric phosphor-screen rings for the two graphite spacings.

(v)

Problem

The first intensity peak occurs at 10.0°. Calculate the accelerating voltage required.

Solution

$$\lambda = d \sin 10^\circ = 2.14 \times 10^{-11}\,\text{m}, \qquad V = \frac{h^2}{2me\lambda^2} = \boxed{3.30 \times 10^3\,\text{V}}$$

The worksheet requests the classical result; the relativistic correction at $\sim 3.3\,\text{kV}$ is small.

(vi)

Problem

Calculate the angles of all other allowed peaks and use them to refine the sketch from the previous part.

Solution

$$\sin\phi_n = n \sin 10^\circ,$$

so only $n = 1, \ldots, 5$ are possible:

n	1	2	3	4	5
ϕ_n	10.0°	20.3°	31.4°	44.0°	60.3°

with symmetric peaks at negative angles. No sixth order exists because $6 \sin 10^\circ > 1$.

Computational link: Task 6 computes allowed orders, scattering angles and ring radii, and can replace the classical wavelength by its relativistic form.

Question 6

Problem

The Lyman series consists of transitions $E_n \to E_1$, where $n \geq 2$. Calculate the wavelength for $E_3 \to E_1$. Repeat for the Balmer transition $E_4 \to E_2$ and the Pfund transition $E_7 \to E_5$, and identify the electromagnetic region containing each line.

Solution

For hydrogen,

$$E_\gamma = 13.6 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \text{eV}, \qquad \lambda = \frac{hc}{E_\gamma}.$$

Thus

Series	Transition	λ	Region
Lyman	$3 \to 1$	$102.6\,\text{nm}$	ultraviolet
Balmer	$4 \to 2$	$486.1\,\text{nm}$	visible
Pfund	$7 \to 5$	$4.65\,\mu\text{m}$	infrared

The ordering follows directly from the decreasing level separations at larger n.

Computational link: Task 5 generates these transitions programmatically over allowed pairs (n_i, n_f), then plots photon energy against wavelength and groups the lines into spectral series.

Quantum Mechanics 2

Question 1

(i)

Problem

The SI second is defined using radiation associated with a transition between two hyperfine levels of the ground state of caesium-133, of frequency 9 192 631 770 Hz. Calculate its wavelength and identify its region of the electromagnetic spectrum.

Solution

$$\lambda = \frac{c}{f} = \boxed{3.261 \times 10^{-2}\,\mathrm{m} = 3.261\,\mathrm{cm}},$$

which is microwave radiation.

(ii)

Problem

Calculate the energy change corresponding to the caesium-133 transition in electron-volts. State whether it is an electronic or nuclear change.

Solution

$$E = hf = 6.09 \times 10^{-24}\,\mathrm{J} = \boxed{3.81 \times 10^{-5}\,\mathrm{eV}}$$

It is a hyperfine transition involving electron–nuclear spin coupling, not a transition between nuclear energy levels.

Computational link: Task 5 uses the same $E = hf = hc/\lambda$ conversion for each discrete atomic transition, although its hydrogenic energy gaps are much larger than this hyperfine splitting.

Question 2

(i)

Problem

No electrons are emitted from a metal until light of wavelength 300 nm or shorter is incident. Use this threshold to calculate the work function in electron-volts.

Solution

$$\phi = \frac{hc}{\lambda_0} = \boxed{4.14\,\mathrm{eV}}$$

Computational link: Task 4 reads this threshold into the work-function parameter used for the stopping-voltage curve.

(ii)

Problem

A 100 W, 200 nm laser is shone on the metal. Assuming that the kinetic-energy power carried by the photoelectron current is 10% of the incident optical power, calculate the resulting current.

Solution

Each electron has

$$K = \frac{hc}{\lambda} - \phi = 2.07\,\text{eV} = 3.31 \times 10^{-19}\,\text{J}.$$

With electron kinetic-power $P_e = 10\,\text{W}$,

$$I = e\frac{P_e}{K} = \boxed{4.83\,\text{A}}$$

This uses the sheet's idealised statement that 10% of the optical power becomes electron kinetic energy and all emitted charge is collected.

Computational link: Task 4 separates energy and rate effects in the same way: wavelength fixes $K_{\max}$, while optical power fixes the available current.

Question 3

Problem

Estimate the de Broglie wavelength of a water molecule at 300 K. Assume $E_\text{k} = \frac{3}{2}k_\text{B}T$, model water as $2 \times {}^1_1\text{H} + {}^{16}_{8}\text{O}$, and approximate its mass by adding proton and neutron masses. Ignore electron masses and nuclear mass deficits.

Solution

Ignoring electrons and binding deficits,

$$m_{\text{H}_2\text{O}} \simeq 10m_p + 8m_n = 3.01 \times 10^{-26}\,\text{kg}.$$

Since $p = \sqrt{2mK} = \sqrt{3mk_BT}$,

$$\lambda = \frac{h}{\sqrt{3mk_BT}} = \boxed{3.43 \times 10^{-11}\,\text{m} = 0.0343\,\text{nm}}$$

This is a representative thermal value; an equilibrium gas has a distribution of momenta.

Computational link: Task 2 can use the thermal momentum scale $\sqrt{3mk_BT}$ to choose physically motivated speeds for the randomly moving small particles, while their directions remain stochastic.

Question 4

(i)

Problem

Calculate the voltage needed to accelerate an electron from rest to $v = c/2$ using classical kinetic energy. Repeat using

$$E_\text{k} = (\gamma - 1)m_ec^2, \qquad \gamma = (1 - v^2/c^2)^{-1/2}.$$

Solution

$$V_\text{cl} = \frac{m_ec^2}{8e} = \boxed{64.0\,\text{kV}}$$

With $\gamma = 2/\sqrt{3} = 1.1547$,

$$V_{\text{rel}} = \frac{(\gamma - 1)m_e c^2}{e} = \boxed{79.1\,\text{kV}}$$

The classical value is about 19% below the relativistic requirement.

Computational link: Task 6's relativistic extension replaces $eV = \frac{1}{2}mv^2$ *with* $eV = (\gamma - 1)mc^2$ *when the voltage is large.*

(ii)

Problem

Calculate the electron's de Broglie wavelength at $v = c/2$ using both classical momentum $p = m_e v$ and relativistic momentum $p = \gamma m_e v$.

Solution

$$\lambda_{\text{cl}} = \frac{h}{m_e v} = \boxed{4.85\,\text{pm}}, \qquad \lambda_{\text{rel}} = \frac{h}{\gamma m_e v} = \boxed{4.20\,\text{pm}}$$

Computational link: Task 6 can toggle between these classical and relativistic wavelengths to show the resulting inward shift of diffraction rings.

(iii)

Problem

A beam of electrons travelling at $c/2$ is diffracted by an atomic lattice through a total scattering angle of $2°$. Assuming a first-order ring, calculate the lattice spacing.

Solution

Bragg's law uses the glancing angle $\theta = 1°$:

$$d = \frac{\lambda_{\text{rel}}}{2\sin 1°} = \boxed{1.20 \times 10^{-10}\,\text{m} = 0.120\,\text{nm}}$$

For this small angle, $2\sin 1° \approx \sin 2°$.

Computational link: Task 6 performs this Bragg-angle calculation for both graphite spacings and converts the scattering angle into screen radius.

Question 5

(i)

Problem

Use the hydrogenic Bohr model to predict the maximum wavelength capable of ionising: (a) hydrogen, $Z = 1$; (b) helium, $Z = 2$; and (c) radon, $Z = 86$.

Solution

For a one-electron ion,

$$E_{\text{ion}} = 13.6Z^2\,\text{eV}, \qquad \lambda_{\max} = \frac{hc}{E_{\text{ion}}}.$$

Therefore

Z	1	2	86
$\lambda_{\max}$	91.2 nm	22.8 nm	0.0123 nm

The last value is in the X-ray range. These are hydrogenic predictions, not accurate first-ionisation energies of neutral multi-electron atoms.

Computational link: Tasks 5 and 10 both use the hydrogenic Z^2 scaling: Task 5 for energy-level transitions and Task 10 for one-electron orbital wavefunctions.

(ii)

Problem

The measured first molar ionisation energies are

Element	First ionisation energy
Hydrogen	$1312.0\,\mathrm{kJ\,mol^{-1}}$
Helium	$2372.3\,\mathrm{kJ\,mol^{-1}}$
Radon	$1037\,\mathrm{kJ\,mol^{-1}}$

Comment on their disagreement with the one-electron Bohr predictions, identify the key differing assumption, and calculate the corresponding threshold photon wavelengths.

Solution

Dividing by N_A gives approximately 13.60, 24.59 and 10.74 eV for H, He and Rn. Hence

	H	He	Rn
$\lambda_{\max}$	91.2 nm	50.4 nm	115 nm

Hydrogen agrees because it is a one-electron atom. In neutral helium and radon, shielding, electron repulsion, outer-shell quantum number and effective nuclear charge invalidate the substitution $E = -13.6Z^2/n^2$ for the first electron removed.

Computational link: Task 10's Z-dependent orbital plots remain hydrogenic; this comparison explains why they should not be presented as exact models of neutral multi-electron atoms.

Question 6

Problem

FAST has diameter 500 m and observes Andromeda at distance 2.0×10^{22} m, tuned to 1420 MHz. Assume isotropic power 8.0×10^{27} W at this frequency. Predict the photon rate received. Compare it with the photons incident on a 24 mm × 36 mm camera sensor exposed to $100\,\mathrm{W\,m^{-2}}$ sunlight for 1/500 s, taking 500 nm as a typical wavelength. How long must FAST observe to receive the same photon number?

Solution

At $r = 2.0 \times 10^{22}$ m, an isotropic luminosity $L = 8.0 \times 10^{27}$ W gives

$$I = \frac{L}{4\pi r^2} = 1.59 \times 10^{-18}\,\mathrm{W\,m^{-2}}.$$

For a 500 m diameter aperture, $A = 1.963 \times 10^5\,\mathrm{m}^2$, so at $f = 1.420\,\mathrm{GHz}$,

$$\dot{N}_{\mathrm{FAST}} = \frac{IA}{hf} = \boxed{3.32 \times 10^{11}\,\mathrm{s}^{-1}}$$

A $24 \times 36\,\mathrm{mm}^2$ sensor receiving $100\,\mathrm{W\,m}^{-2}$ for $1/500\,\mathrm{s}$ receives $1.728 \times 10^{-4}\,\mathrm{J}$. At 500 nm,

$$N_{\mathrm{camera}} = \frac{E}{hc/\lambda} = \boxed{4.35 \times 10^{14}}$$

Thus

$$t = \frac{N_{\mathrm{camera}}}{\dot{N}_{\mathrm{FAST}}} = \boxed{1.31 \times 10^3\,\mathrm{s} = 21.8\,\mathrm{min}}$$

Question 7

(i)

Problem

A solar sail of area A reflects incident light of intensity I with 100% efficiency. If the light has average wavelength λ, use photon energy and the de Broglie relation to derive the impulse delivered to the sail per second.

Solution

The photon rate is $\dot{N} = IA/(hc/\lambda)$, while each reflected photon transfers $2h/\lambda$. Therefore

$$\boxed{F = \dot{p} = \frac{2IA}{c}}$$

For absorption, the factor 2 is absent.

(ii)

Problem

A spacecraft of total mass m, including the sail, uses the reflected radiation for propulsion. Derive its acceleration, neglecting relativistic effects.

Solution

$$\boxed{a = \frac{2IA}{mc}}$$

(iii)

Problem

Light of wavelength 500 nm has intensity $1.0\,\mathrm{kW\,m}^{-2}$. Calculate the acceleration of a 10-tonne spacecraft with sail area $100\,\mathrm{km}^2$, and the time needed to reach $0.01c$ under constant classical acceleration.

Solution

With $A = 10^8\,\mathrm{m}^2$ and $m = 10^4\,\mathrm{kg}$,

$$a = \boxed{6.67 \times 10^{-2}\,\mathrm{m\,s}^{-2}}$$

At constant intensity and classical acceleration,

$$t = \frac{0.01c}{a} = \boxed{4.50 \times 10^7\,\mathrm{s} \approx 520\,\mathrm{d} \approx 1.43\,\mathrm{yr}}$$

For a stellar source, the more serious limitation is $I \propto r^{-2}$, not relativity at $0.01c$.

Question 8

(i)

Problem

The maximum solar intensity near the ISS is $1361\,\mathrm{W\,m^{-2}}$. Estimate the Sun's luminosity. Assuming this average power for a 10^{10}-year main-sequence lifetime, use $E = mc^2$ to estimate the percentage decrease in solar mass.

Solution

At $r = 1\,\mathrm{AU} = 1.496 \times 10^{11}\,\mathrm{m}$,

$$L_\odot = 4\pi r^2 I = \boxed{3.83 \times 10^{26}\,\mathrm{W}}$$

Over $3.156 \times 10^{17}\,\mathrm{s}$, $E = 1.21 \times 10^{44}\,\mathrm{J}$, so

$$\Delta m = \frac{E}{c^2} = 1.34 \times 10^{27}\,\mathrm{kg}, \qquad \frac{\Delta m}{M_\odot} \times 100\% = \boxed{0.0675\%}$$

This excludes solar-wind loss and assumes constant luminosity.

(ii)

Problem

Sirius has $T_S = 9940\,\mathrm{K}$, $R_S = 1.711 R_\odot$ and $M_S = 2.02 M_\odot$. Calculate: (a) the intensity at one astronomical unit; (b) its luminosity; and (c) its main-sequence lifetime if it converts the same fraction of its mass into radiation as the Sun.

Solution

$$\frac{L_S}{L_\odot} = (1.711)^2 \left(\frac{9940}{5778}\right)^4 = 25.6.$$

Hence, at one AU,

$$I_S = \boxed{3.49 \times 10^4\,\mathrm{W\,m^{-2}}}, \qquad L_S = \boxed{9.82 \times 10^{27}\,\mathrm{W}}$$

If the same fraction of stellar mass is radiated,

$$t_S = t_\odot \frac{M_S/M_\odot}{L_S/L_\odot} = \boxed{7.88 \times 10^8\,\mathrm{yr}}$$

Computational link: Task 3's black-body model reproduces this $L \propto R^2T^4$ scaling and shows how the hotter star's spectrum shifts to shorter wavelengths.

(iii)

Problem

A "DogStar" heater has power equal to that intercepted by a $10\,\mathrm{m}^2$ panel at one astronomical unit from Sirius. Using $c_w = 4181\,\mathrm{J\,kg^{-1}\,K^{-1}}$, estimate the ideal time to heat one litre of water from 20°C to boiling.

Solution

$$Q = mc\Delta T = (1)(4181)(80) = 3.345 \times 10^5\,\mathrm{J},$$

$$P = I_S A = 3.49 \times 10^5\,\mathrm{W}, \qquad \boxed{t = Q/P = 0.959\,\mathrm{s}}$$

This neglects the vessel and all losses.

Quantum Mechanics 3

Question 1

(i)

Problem

An electron travels at speed $c/137$. Calculate its de Broglie wavelength.

Solution

At this speed $\gamma - 1 \ll 1$, so

$$\lambda = \frac{h}{m_e v} = \frac{137h}{m_e c} = \boxed{3.33 \times 10^{-10}\,\mathrm{m}}$$

Computational link: Task 6 uses this de Broglie relation to convert electron momentum into the wavelength controlling diffraction-ring positions.

(ii)

Problem

An electron and a neutron have equal de Broglie wavelengths. Assuming classical physics, find the ratio of their speeds and the ratio of their kinetic energies.

Solution

Equal λ means equal momentum. Thus

$$\frac{v_e}{v_n} = \frac{m_n}{m_e} = \boxed{1.84 \times 10^3}$$

Since $K = p^2/(2m)$,

$$\frac{K_e}{K_n} = \frac{m_n}{m_e} = \boxed{1.84 \times 10^3}$$

(iii)

Problem

Use $\Delta x\,\Delta p \approx \hbar/2$, $\Delta E\,\Delta t \approx \hbar/2$, $\Delta p \approx mc$ and $\Delta E \approx mc^2$ to estimate the range Δx and lifetime Δt of a Z boson of mass $91.2\,\mathrm{GeV}/c^2$.

Solution

Using $\Delta x \Delta p \approx \hbar/2$, $\Delta E \Delta t \approx \hbar/2$, $\Delta p \sim mc$ and $\Delta E \sim mc^2$,

$$\Delta x \sim \frac{\hbar}{2mc} = \boxed{1.08 \times 10^{-18}\,\mathrm{m}}, \qquad \Delta t \sim \frac{\hbar}{2mc^2} = \boxed{3.61 \times 10^{-27}\,\mathrm{s}}$$

These are uncertainty-based scales, not precision lifetime measurements.

Computational link: Task 7 evaluates $\Delta x\,\Delta p$ explicitly for box eigenstates; this estimate shows the same uncertainty principle acting on much shorter-lived relativistic particles.

(iv)

Problem

Show that

$$\psi(x,t) = A\sin(kx - \omega t)$$

is a solution of the one-dimensional wave equation.

Solution

$$\frac{\partial^2 \psi}{\partial x^2} = -k^2\psi, \qquad \frac{\partial^2 \psi}{\partial t^2} = -\omega^2\psi.$$

For $\omega = ck$,

$$\boxed{\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 \psi}{\partial t^2}}$$

(v)

Problem

Explain wavefunction collapse using electron diffraction in the double-slit experiment.

Solution

Before detection, amplitudes from both slits superpose:

$$\psi = \psi_1 + \psi_2, \qquad |\psi|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\,\mathrm{Re}(\psi_1^*\psi_2).$$

The cross term produces interference. A screen measurement yields one localised impact: the state is updated to one compatible with that outcome. A which-path measurement destroys coherence between the alternatives, removing the interference pattern. The propagation is wave-like; each detection is localised.

Computational link: Task 6 uses precisely this separation between wave-like propagation and localised screen impacts when rendering electron diffraction rings.

(vi)

Problem

Show that the classical kinetic energy of a particle can be written as

$$E_\mathrm{k} = \frac{p^2}{2m}.$$

Solution

Classically,

$$p = mv \quad \Rightarrow \quad v = \frac{p}{m}.$$

Substituting into $K = mv^2/2$,

$$K = \frac{1}{2}m\left(\frac{p}{m}\right)^2 = \boxed{\frac{p^2}{2m}}$$

Combining this with $p = h/\lambda$ gives $K = h^2/(2m\lambda^2)$.

(vii)

Problem

Neutrons are scattered by a lattice of spacing 2.15×10^{-10} m. If the angular spacing between the central and first-order maxima is 30°, calculate their de Broglie wavelength, kinetic energy in electron-volts and speed, assuming classical mechanics.

Solution

For first order,

$$\lambda = s\sin 30^\circ = \boxed{1.075 \times 10^{-10}\,\mathrm{m}}$$

Then

$$p = \frac{h}{\lambda} = 6.16 \times 10^{-24}\,\mathrm{kg\,m\,s^{-1}},$$

$$K = \frac{p^2}{2m_n} = \boxed{7.07 \times 10^{-2}\,\mathrm{eV}}, \qquad v = \frac{p}{m_n} = \boxed{3.68 \times 10^3\,\mathrm{m\,s^{-1}}}$$

Computational link: Task 6 applies the same lattice-diffraction workflow to electrons, with accelerating voltage replacing the neutron speed as the control parameter.

(viii)

Problem

Silver has work function 4.3 eV, and the maximum kinetic energy of emitted photoelectrons is 5.1 eV. Calculate the ultraviolet wavelength. Then calculate the kinetic energy of an electron having the same de Broglie wavelength.

Solution

$$E_\gamma = 9.4\,\mathrm{eV}, \qquad \lambda = \frac{hc}{E_\gamma} = \boxed{132\,\mathrm{nm}}$$

For an electron with this de Broglie wavelength,

$$K = \frac{h^2}{2m_e\lambda^2} = \boxed{8.6 \times 10^{-5}\,\mathrm{eV}}$$

A photon and a massive non-relativistic particle obey different energy–momentum relations.

Computational link: This combines Task 4's photoelectric energy balance with the de Broglie conversion used in Task 6.

(ix)

Problem

Diffraction limits microscope resolution to a length scale of order the wavelength. Calculate the energy in electron-volts of electrons corresponding to a resolution limit of 0.42 nm.

Solution

Taking $\lambda \sim \Delta x$,

$$K = \frac{h^2}{2m_e\lambda^2} = \boxed{8.5\,\text{eV}}$$

Real microscopes use much higher voltages because resolution is also limited by lens aberrations and practical optics.

Computational link: Task 6's inverse relationship between wavelength and momentum also explains why higher accelerating voltages improve electron-beam spatial resolution.

(x)

Problem

A 5 MeV electron is emitted in beta decay. Use the uncertainty principle to estimate a lower limit for the interaction time and the spatial extent of the process.

Solution

With $\Delta E \sim 5\,\text{MeV}$,

$$\Delta t \gtrsim \frac{\hbar}{2\Delta E} = \boxed{6.59 \times 10^{-23}\,\text{s}}$$

The electron is relativistic, so $\Delta p \sim \Delta E/c$, giving

$$\Delta x \gtrsim \frac{\hbar c}{2\Delta E} = \boxed{1.97 \times 10^{-14}\,\text{m}}$$

Computational link: Task 7's uncertainty calculation is an exact eigenstate analysis; this is the complementary order-of-magnitude use of the energy–time and position–momentum relations.

Question 2

A particle is confined to $0 < x < a$, with $V = 0$ inside and $V = \infty$ outside.

(i)

Problem

A particle of mass m is confined to an infinite one-dimensional box over $0 \le x \le a$. Explain why

$$\psi_n(x,t) = A_n \sin\left(\frac{n\pi x}{a}\right)\cos(\omega t)$$

is a sensible standing-wave form and state what type of number n must be.

Solution

The walls require $\psi(0) = \psi(a) = 0$. Hence

$$\sin(n\pi) = 0 \quad \Rightarrow \quad n = 1, 2, 3, \ldots, \qquad \lambda_n = \frac{2a}{n}.$$

The value $n = 0$ gives the zero function and is not a physical state.

(ii)

Problem

Show that the spatial part is consistent with the time-independent Schrödinger equation and obtain $E_n = E(n, a)$. Evaluate the proton ground-state energy in MeV for $a = 0.47 \times 10^{-15}$ m, and compare it with $m_p c^2$.

Solution

Inside the well,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi, \qquad \frac{d^2\psi_n}{dx^2} = -\left(\frac{n\pi}{a}\right)^2 \psi_n.$$

Therefore

$$\boxed{E_n = \frac{n^2\pi^2\hbar^2}{2ma^2} = \frac{n^2h^2}{8ma^2}}$$

For a proton,

$$E_1 = \boxed{9.27 \times 10^2\,\text{MeV}}, \qquad m_pc^2 = 938.3\,\text{MeV}.$$

Because the predicted kinetic energy is comparable to the rest energy, the non-relativistic Schrödinger model is no longer quantitatively reliable at this width.

Computational link: Task 7 plots this quadratic $E_n \propto n^2$ spectrum and exposes how changing the box width rescales every energy as a^{-2}.

(iii)

Problem

Explain why the time factor should be $\exp(-\mathrm{i}E_nt/\hbar)$ rather than $\cos\omega t$, and use the Born interpretation to determine the normalisation constant A_n.

Solution

For an energy eigenstate,

$$i\hbar\frac{dT}{dt} = E_nT \quad \Rightarrow \quad T(t) = \mathrm{e}^{-iE_nt/\hbar}.$$

Normalisation requires

$$1 = |A_n|^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = |A_n|^2\frac{a}{2},$$

so

$$\boxed{\psi_n(x,t) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi x}{a}\right)\mathrm{e}^{-iE_nt/\hbar}}$$

Its probability density is time-independent.

Computational link: Task 7 plots the normalised $|\psi_n|^2$ profiles; the phase factor disappears from the probability density exactly as shown here.

(iv)

Problem

Explain why $\langle p \rangle = 0$ and show that

$$\langle p^2 \rangle = \frac{n^2\pi^2\hbar^2}{a^2}.$$

Solution

Using $\hat{p} = -i\hbar d/dx$,

$$\langle p \rangle = -\frac{i\hbar}{2}[\psi_n^2]_0^a = 0.$$

Also,

$$\hat{p}^2\psi_n = -\hbar^2\psi_n'' = \frac{n^2\pi^2\hbar^2}{a^2}\psi_n,$$

so

$$\boxed{\left\langle p^2 \right\rangle = \frac{n^2\pi^2\hbar^2}{a^2}}, \qquad \boxed{\Delta p = \frac{n\pi\hbar}{a}}$$

The zero mean reflects equal $+p$ and $-p$ components in a standing wave.

Computational link: Task 7 computes this momentum spread for each slider-selected n, providing one half of the uncertainty-product verification.

(v)

Problem

Prove that

$$\langle x \rangle = \frac{a}{2}, \qquad \langle x^2 \rangle = a^2\left(\frac{1}{3} - \frac{1}{2n^2\pi^2}\right).$$

Solution

Using the normalised density,

$$\langle x \rangle = \frac{2}{a}\int_0^a x\sin^2\left(\frac{n\pi x}{a}\right)dx.$$

With $\sin^2 u = (1-\cos 2u)/2$, the cosine integral vanishes at the boundaries, so

$$\langle x \rangle = \frac{1}{a}\int_0^a x\,dx = \boxed{\frac{a}{2}}$$

Similarly,

$$\langle x^2 \rangle = \frac{1}{a}\int_0^a x^2\,dx - \frac{1}{a}\int_0^a x^2\cos\left(\frac{2n\pi x}{a}\right)dx.$$

Writing $k = 2n\pi/a$, integration by parts gives

$$\int_0^a x^2\cos(kx)dx = \frac{2a}{k^2} = \frac{a^3}{2n^2\pi^2}.$$

Therefore

$$\boxed{\langle x^2 \rangle = a^2\left(\frac{1}{3} - \frac{1}{2n^2\pi^2}\right)}$$

Finally,

$$(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2 = a^2\left(\frac{1}{12} - \frac{1}{2n^2\pi^2}\right),$$

so

$$\boxed{\Delta x = a\sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}}}$$

Computational link: Task 7 evaluates these position moments numerically or analytically and combines them with Δp in the extension.

(vi)

Problem

Hence show that

$$\Delta x\,\Delta p = \frac{\hbar}{2}\sqrt{\frac{n^2\pi^2}{3} - 2},$$

and verify that every allowed state satisfies Heisenberg's uncertainty principle.

Solution

Substituting the results from parts (iv) and (v),

$$\begin{aligned}\Delta x\,\Delta p &= a\sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}}\left(\frac{n\pi\hbar}{a}\right)\\ &= \frac{\hbar}{2}\sqrt{\frac{n^2\pi^2}{3} - 2}.\end{aligned}$$

For $n \geq 1$, $n^2\pi^2 \geq \pi^2 > 9$. Hence the square-root factor is greater than one and

$$\boxed{\Delta x\,\Delta p > \frac{\hbar}{2}}$$

For the ground state the product is $0.568\hbar$, so the inequality is satisfied but not saturated.

Computational link: Task 7 visualises the same E_n, nodes and $|\psi_n|^2$; the analysis above verifies the uncertainty relation for every plotted state.

Question 3

Problem

Two vertically polarised entangled photons travel to detectors A and B, whose axes are at $\theta = -45°$ and $\phi = 30°$. Using Malus-law probabilities, calculate the probability of different X, Y outcomes first if the measurements are independent, and then using the Copenhagen interpretation with detector A measuring first.

Solution

For independent vertically polarised photons,

$$P(X_A) = \cos^2\theta = \frac{1}{2}, \qquad P(X_B) = \cos^2\phi = \frac{3}{4}.$$

Thus

$$P_{\text{diff}}^{\text{ind}} = P(X_A)P(Y_B) + P(Y_A)P(X_B) = \boxed{\frac{1}{2}}$$

For the perfectly correlated entangled state assumed by the worksheet, the second result is conditioned on the first and depends only on the relative analyser angle:

$$P_{\rm diff}^{\rm ent} = \sin^2(\theta - \phi) = \sin^2(75^\circ) = \boxed{0.933}$$

The phrase "two vertically polarised entangled photons" should not be read as the separable state $|V\rangle|V\rangle$; the calculation assumes an entangled state with equal outcomes for aligned analysers.

Computational link: Task 8 is a visual calculator for these two mismatch models, varying both analyser angles and displaying the independent and entangled probabilities side by side.

High-Energy and Particle Physics

Question 1

(i)

Problem

For beta-minus decay, the vertices are $d \to u + W^-$ and $W^- \to e^- + \bar{\nu}_e$. Show that electric charge, baryon number and lepton number are conserved at each vertex.

Solution

At the quark vertex,

$$Q:\ -\tfrac{1}{3} = \tfrac{2}{3} - 1, \qquad B:\ \tfrac{1}{3} = \tfrac{1}{3} + 0, \qquad L_e:\ 0 = 0 + 0.$$

At the leptonic vertex,

$$Q:\ -1 = -1 + 0, \qquad B:\ 0 = 0 + 0, \qquad L_e:\ 0 = 1 - 1.$$

Thus charge, baryon number and electron-lepton number are conserved. The antineutrino is required for lepton-number conservation.

(ii)

Problem

Sketch a Feynman diagram for beta-plus decay and verify charge, baryon-number and lepton-number conservation at each interaction vertex.

Solution

At quark level,

$$u \to d + W^+, \qquad W^+ \to e^+ + \nu_e.$$

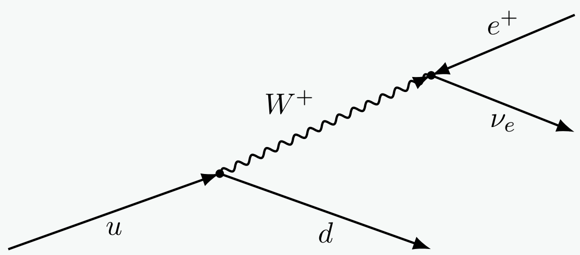

Figure 4: Quark-level beta-plus decay; antifermion arrows point towards the vertex.

The checks are
$$Q:\ \tfrac{2}{3} = -\tfrac{1}{3} + 1, \quad B:\ \tfrac{1}{3} = \tfrac{1}{3}, \quad L_e:\ 0 = -1 + 1.$$

(iii)

Problem

Using quark content, show that total charge and baryon number are conserved in
$$\pi^- + p \to n + \pi^- + \pi^+.$$

Solution

Using $\pi^- = d\bar{u}$, $p = uud$, $n = udd$, and $\pi^+ = u\bar{d}$,
$$Q_i = -1 + 1 = 0 = 0 - 1 + 1 = Q_f,$$
$$B_i = 0 + 1 = 1 = 1 + 0 + 0 = B_f.$$

(iv)

Problem

A kaon decays by $K^+ \to \pi^+ + \pi^0$. Use the quark compositions of the mesons to show that charge is conserved.

Solution

$$K^+ = u\bar{s}: \quad Q = \tfrac{2}{3} + \tfrac{1}{3} = 1,$$
$$\pi^+ = u\bar{d}:\ Q = 1, \qquad \pi^0:\ Q = 0.$$

Hence $1 = 1 + 0$. More precisely, $\pi^0 = (u\bar{u} - d\bar{d})/\sqrt{2}$, and either component is neutral.

(v)

Problem

Calculate the worksheet's "quark binding energy" for (a) K^+ and (b) π^+, given $m_{K^+} = 0.52616 m_p$ and $m_{\pi^+} = 0.14875 m_p$. Use MeV/c^2 as the mass unit.

Solution

Define
$$B_q = \left(\sum m_{\text{current quarks}} - m_{\text{hadron}}\right) c^2.$$

Using $m_p c^2 = 938.27\,\mathrm{MeV}$, $m_u = 2.4$, $m_d = 4.8$, $m_s = 95\,\mathrm{MeV}/c^2$,
$$B_{K^+} = 2.4 + 95 - 0.52616(938.27) = \boxed{-396.3\,\mathrm{MeV}},$$
$$B_{\pi^+} = 2.4 + 4.8 - 0.14875(938.27) = \boxed{-132.4\,\mathrm{MeV}}$$

These negative values are not ordinary nuclear binding energies: current quark masses omit the large gluon-field, confinement and quark kinetic contributions to hadron mass.

(vi)

Problem

Sketch the Feynman diagram for

$$\Sigma^- \to n + e^- + \bar{\nu}_e$$

and perform the same conservation-law checks as for beta decay.

Solution

Because $\Sigma^- = dds$ and $n = udd$,

$$s \to u + W^-, \qquad W^- \to e^- + \bar{\nu}_e.$$

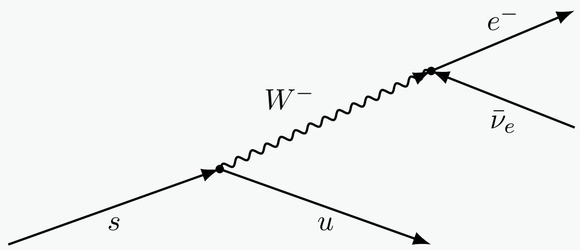

Figure 5: Weak decay of the strange quark in Σ^-.

At the first vertex,

$$Q: \ -\tfrac{1}{3} = \tfrac{2}{3} - 1, \quad B: \ \tfrac{1}{3} = \tfrac{1}{3}, \quad L_e: \ 0 = 0.$$

At the second, $Q: -1 = -1 + 0$, $B: 0 = 0$, $L_e: 0 = 1 - 1$. Strangeness changes from -1 to 0, as allowed in a weak interaction.

(vii)

Problem

Briefly summarise how cloud chambers and bubble chambers reveal the tracks of charged particles.

Solution

A cloud chamber contains supersaturated vapour; ions left by a charged particle seed condensation droplets. A bubble chamber contains a superheated liquid; ions seed bubbles after pressure reduction. Track density, curvature in a magnetic field, branching and vertices reveal ionisation, charge sign, momentum-to-charge ratio and interactions. Bubble chambers provide a denser target.

(viii)

Problem

A gamma ray may create a $\mu^-\mu^+$ pair. Calculate the threshold energy in joules and the maximum photon wavelength, given $m_\mu c^2 = 105.66\,\text{MeV}$.

Solution

Ignoring recoil,

$$E_{\text{min}} = 2m_\mu c^2 = 211.32\,\text{MeV} = \boxed{3.385 \times 10^{-11}\,\text{J}},$$

$$\lambda_{\max} = \frac{hc}{E_{\min}} = \boxed{5.87 \times 10^{-15}\,\mathrm{m}}$$

A lone photon cannot create a pair in vacuum while conserving four-momentum; a nucleus or other body must recoil, making the exact threshold slightly higher.

(ix)

Problem

A gamma ray produces an electron–positron pair, with each particle moving at $0.8c$. Using relativistic total energy, determine the gamma-ray energy in MeV.

Solution

$$\gamma = (1 - 0.8^2)^{-1/2} = \frac{5}{3},$$

so

$$E_\gamma = 2\gamma m_e c^2 = 2\left(\frac{5}{3}\right)(0.511) = \boxed{1.70\,\mathrm{MeV}}$$

(x)

Problem

Explain, with a suitable diagram, how a single high-energy electron can generate an electron–positron shower through bremsstrahlung and pair production, allowing the original electron to be detected.

Solution

A high-energy electron emits bremsstrahlung when deflected in a nuclear field. The photon can convert into e^-e^+ near another nucleus; the secondaries radiate further photons, causing a cascade until energies are too low for efficient bremsstrahlung or pair production.

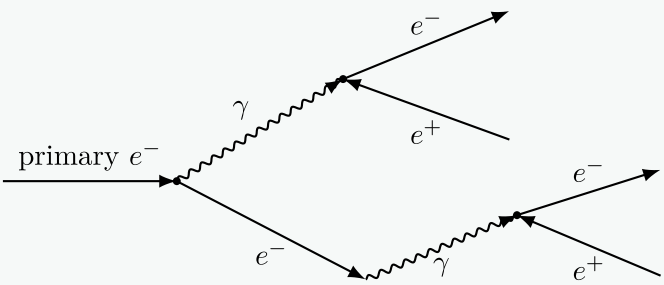

Figure 6: Repeated bremsstrahlung and pair production form a shower.

The detector measures the combined ionisation, scintillation or deposited energy, amplifying the original particle's signature.

(xi)

Problem

For the u, d, s quarks, evaluate

$$\frac{9(\sqrt{m_u} + \sqrt{m_d} + \sqrt{m_s})^2}{m_u + m_d + m_s}$$

and identify the nearest integer k.

Solution

$$k = \frac{9(\sqrt{2.4} + \sqrt{4.8} + \sqrt{95})^2}{2.4 + 4.8 + 95} = 16.02,$$

so $\boxed{k = 16}$ to the nearest integer. Quark masses are scale- and scheme-dependent.

Question 2

(i)

Problem

Strangeness is minus the number of strange quarks, while an antistrange quark contributes +1. In

$$K^- + p \to \Omega^- + K^+ + K^0,$$

determine the quark content of K^0 so that charge, baryon number and strangeness are conserved.

Solution

Initially $Q = 0$, $B = 1$, $S = -1$. The known final particles give

$$\Omega^- = sss: \ (Q, B, S) = (-1, 1, -3),$$

$$K^+ = u\bar{s}: \ (Q, B, S) = (+1, 0, +1).$$

Therefore K^0 must have $(Q, B, S) = (0, 0, +1)$, so

$$\boxed{K^0 = d\bar{s}}$$

(ii)

Problem

The weak decay

$$\Omega^- \to \Xi^0 + \pi^-$$

occurs. Draw a Feynman diagram, verify charge and baryon-number conservation at each vertex, and show that strangeness is not conserved.

Solution

With $\Omega^- = sss$, $\Xi^0 = uss$ and $\pi^- = d\bar{u}$,

$$s \to u + W^-; \qquad W^- \to d + \bar{u},$$

while two s quarks are spectators.

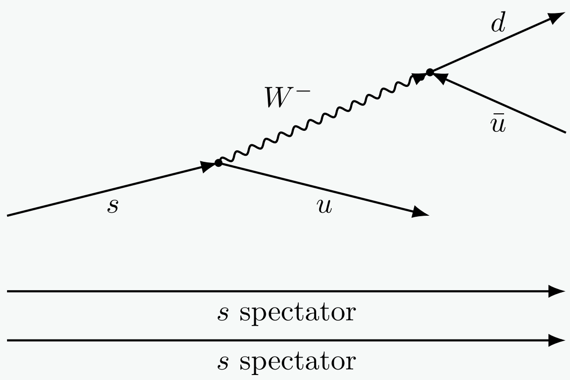

Figure 7: Weak decay $\Omega^- \to \Xi^0 + \pi^-$.

At each vertex, charge and baryon number balance. Overall,

$$Q : -1 = 0 - 1, \quad B : 1 = 1 + 0, \quad S : -3 \to -2,$$

so $\boxed{\Delta S = +1}$, permitted by the weak interaction.

Question 3

(i)

Problem

A tau lepton decays through

$$\tau^- \to \pi^- + \nu_\tau.$$

Draw an appropriate weak-interaction Feynman diagram.

Solution

$$\tau^- \to \nu_\tau + W^-; \qquad W^- \to d + \bar{u}; \qquad \pi^- = d\bar{u}.$$

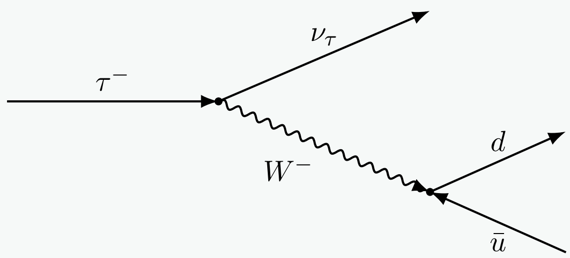

Figure 8: Weak hadronic tau decay.

(ii)

Problem

Explain which conservation law would be violated if the tau neutrino were absent.

Solution

Tau-lepton number would change from +1 to 0. The neutrino is also required by energy–momentum conservation.

(iii)

Problem

A τ^- of rest energy 1776.86 MeV moves at $0.2c$. Calculate its kinetic energy in MeV.

Solution

$$\gamma = (1 - 0.2^2)^{-1/2} = 1.02062,$$

$$K = (\gamma - 1)m_\tau c^2 = (0.02062)(1776.86) = \boxed{36.6\,\text{MeV}}$$

The total energy is 1813.5 MeV.

(iv)

Problem

The pion produced in the decay moves at $0.99c$, and $m_{\pi^-} = 0.14875m_p$. Using the energy balance specified in the sheet, calculate the tau-neutrino energy in MeV.

Solution

$$m_\pi c^2 = 0.14875(938.27) = 139.57\,\text{MeV}, \quad \gamma_\pi = 7.0888,$$

$$E_\pi = 989.4\,\text{MeV}, \qquad E_{\nu_\tau} = 1813.5 - 989.4 = \boxed{824\,\text{MeV}}$$

The separately specified speeds are not guaranteed to satisfy the full vector momentum constraints of a two-body decay; this is the energy-balance result requested by the sheet.

(v)

Problem

Using $E = \gamma mc^2$ and $p = \gamma mu$, show that

$$E^2 - p^2c^2 = m^2c^4.$$

If the tau-neutrino mass is negligible, calculate its momentum.

Solution

With $E = \gamma mc^2$ and $p = \gamma mu$,

$$E^2 - p^2c^2 = \gamma^2 m^2 c^4 \left(1 - \frac{u^2}{c^2}\right) = \boxed{m^2c^4}$$

For negligible neutrino mass, $E = pc$, so

$$\boxed{p_{\nu_\tau} = 824\,\text{MeV}/c = 4.40 \times 10^{-19}\,\text{kg m s}^{-1}}$$

(vi)

Problem

Evaluate the Koide-type expressions

$$\frac{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}{m_e + m_\mu + m_\tau}$$

and
$$\frac{(\sqrt{m_c}+\sqrt{m_b}+\sqrt{m_t})^2}{m_c+m_b+m_t}.$$

Solution

Using masses in a common unit,
$$\frac{(\sqrt{m_e}+\sqrt{m_\mu}+\sqrt{m_\tau})^2}{m_e+m_\mu+m_\tau}=\boxed{1.50002},$$
$$\frac{(\sqrt{m_c}+\sqrt{m_b}+\sqrt{m_t})^2}{m_c+m_b+m_t}=\boxed{1.4945}$$
The charged-lepton value is extremely close to $3/2$; quark masses depend on renormalisation convention and scale.

Question 4

(i)

Problem

Using the energy–momentum invariant for a massless photon and Planck's law, show that
$$p_\lambda=\frac{h}{\lambda}.$$

Solution

For a photon, $m=0$, so $E=pc$. Combining with $E=hf=hc/\lambda$ gives
$$\boxed{p_\lambda=\frac{h}{\lambda}}$$

(ii)

Problem

For an incident photon momentum p_λ, scattered photon momentum $p_{\lambda'}$, scattering angle θ and electron recoil momentum p_e, use momentum conservation and the cosine rule to show
$$p_e^2=p_\lambda^2+p_{\lambda'}^2-2p_\lambda p_{\lambda'}\cos\theta.$$

Solution

Since $\mathbf{p}_e=\mathbf{p}_\lambda-\mathbf{p}_{\lambda'}$,
$$\boxed{p_e^2=p_\lambda^2+p_{\lambda'}^2-2p_\lambda p_{\lambda'}\cos\theta}$$

(iii)

Problem

Use energy conservation, retaining the rest energy of the initially stationary electron, to show
$$p_e^2=(p_\lambda-p_{\lambda'})^2+2m_ec(p_\lambda-p_{\lambda'}).$$

Solution

$$p_\lambda c + m_e c^2 = p_{\lambda'} c + \sqrt{p_e^2 c^2 + m_e^2 c^4}.$$

Isolating the square root and squaring gives

$$\boxed{p_e^2 = (p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'})}$$

(iv)

Problem

Eliminate p_e^2 between the previous two expressions and prove the Compton-shift formula

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta).$$

Solution

Equating the two expressions and cancelling common terms gives

$$p_\lambda p_{\lambda'}(1 - \cos\theta) = m_e c(p_\lambda - p_{\lambda'}).$$

Using $p = h/\lambda$,

$$\boxed{\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)}$$

Computational link: This Compton-shift equation is the central wavelength update implemented in Task 9 for every photon scattering angle.

(v)

Problem

At what angle is the wavelength shift greatest? Calculate the incident X-ray energy in keV when the maximum shift is 10% of the incident wavelength.

Solution

The maximum occurs at backscattering, $\theta = 180^\circ$, so $\Delta\lambda_{\max} = 2h/(m_e c)$. Requiring $\Delta\lambda_{\max} = 0.1\lambda$ gives

$$E = \frac{hc}{\lambda} = \frac{m_e c^2}{20} = \boxed{25.6\,\text{keV}}$$

(vi)

Problem

Resolve momentum to derive the recoil angle

$$\tan\phi = \frac{\sin\theta}{1 + \dfrac{h}{m_e c\lambda}(1 - \cos\theta) - \cos\theta},$$

and show that

$$v = c\left[1 + \left(\frac{m_e c}{p_e}\right)^2\right]^{-1/2},$$

with p_e given by the energy-conservation expression.

Solution

Resolving momentum,

$$p_e \sin\phi = p_{\lambda'} \sin\theta, \quad p_e \cos\phi = p_\lambda - p_{\lambda'} \cos\theta.$$

Therefore

$$\boxed{\tan\phi = \frac{\sin\theta}{1 + \frac{h}{m_e c \lambda}(1-\cos\theta) - \cos\theta}}$$

With

$$p_e = \sqrt{(p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'})},$$

relativistic momentum gives

$$\boxed{v = c\left[1 + \left(\frac{m_e c}{p_e}\right)^2\right]^{-1/2}}$$

Computational link: Task 9 evaluates these expressions over $0 \le \theta \le \pi$ to plot electron recoil angle and relativistic recoil speed.

(vii)

Problem

Explain why photon backscattering produces the maximum electron recoil and show

$$v_{\max} = c\sqrt{1 - \left(\frac{m_e c^2}{m_e c^2 + hc/\lambda - hc/\lambda'}\right)^2}, \qquad \lambda' = \lambda + \frac{2h}{m_e c}.$$

Solution

Backscattering maximises λ', minimises the scattered photon energy and therefore maximises electron energy. With

$$\lambda' = \lambda + \frac{2h}{m_e c},$$

$$E_e = m_e c^2 + \frac{hc}{\lambda} - \frac{hc}{\lambda'},$$

and $v = c\sqrt{1-\gamma^{-2}}$,

$$\boxed{v_{\max} = c\sqrt{1 - \left(\frac{m_e c^2}{m_e c^2 + hc/\lambda - hc/\lambda'}\right)^2}}$$

Computational link: Task 9's curves should peak at backscattering, giving a useful endpoint check on the numerical implementation.

(viii)

Problem

Generate graphs of $\Delta\lambda/\lambda$ and v/c against photon-scattering angle θ for incident X-ray energies $50, 100, 200, 500$ and $1000\,\text{keV}$.

Solution

For each incident energy E, compute $\lambda = hc/E$, then evaluate $\lambda'(\theta)$, $\Delta\lambda/\lambda$, p_e and v/c for $0 \le \theta \le \pi$. Both quantities vanish at $0°$ and increase to maxima at $180°$. Since the absolute shift is energy-independent but $\lambda \propto E^{-1}$, the fractional shift grows with photon energy.

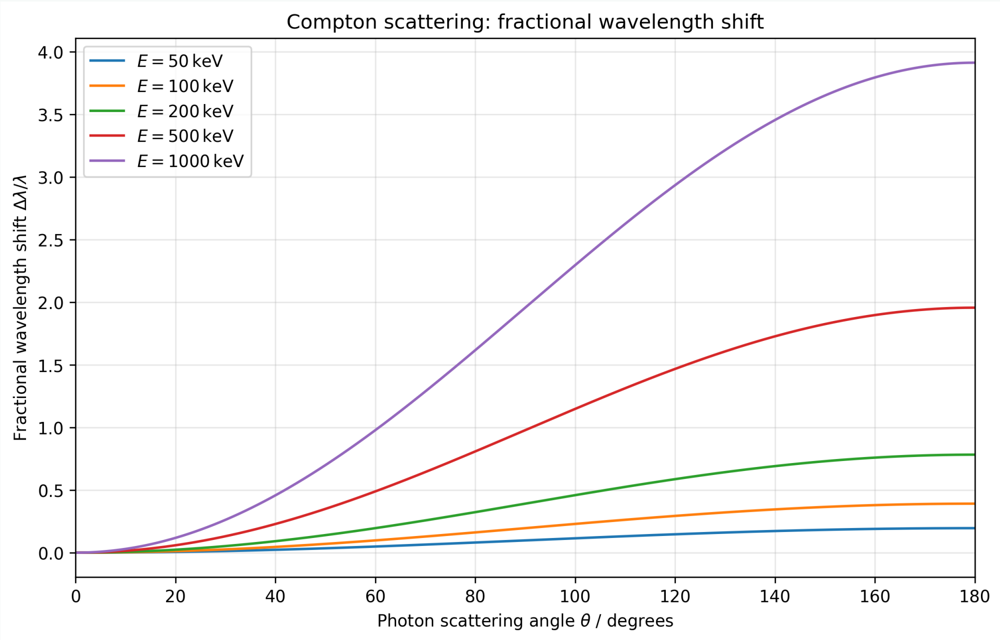

Figure 9: Fractional Compton wavelength shift for the five incident energies.

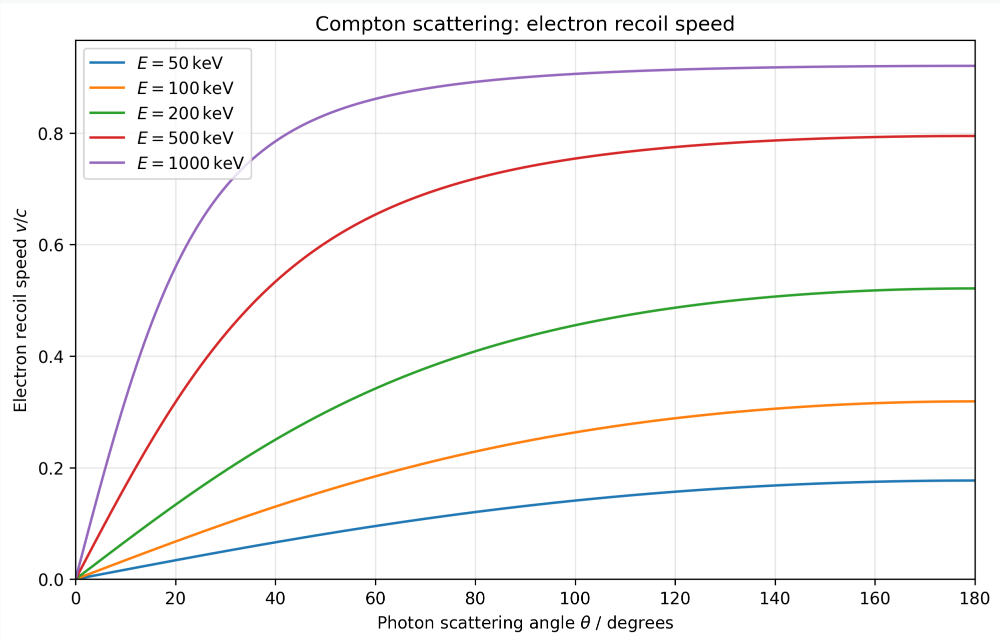

Figure 10: Relativistic electron recoil speed for the five incident energies.

Computational link: These are the core outputs of Task 9.

Nuclear Physics

Question 1

(i)

Problem

Calculate the mass of ${}^{92}_{36}\mathrm{Kr}$ in kilograms and atomic mass units if its binding energy is 764.77 MeV.

Solution

The nucleus contains 36 protons and 56 neutrons:

$$M = 36m_p + 56m_n - \frac{B}{c^2} = \boxed{1.526 \times 10^{-25}\,\mathrm{kg}} = \boxed{91.93\,\mathrm{u}}$$

This is a nuclear mass; electron masses are neglected consistently with the sheet.

(ii)

Problem

The binding energy of ${}^{141}_{56}\mathrm{Ba}$ is 1145.36 MeV. Determine the binding energy of uranium-235 if

$${}^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \to {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3{}^{1}_{0}\mathrm{n}$$

releases 173.28 MeV.

Solution

Free neutrons have zero nuclear binding energy, so

$$173.28 = B_{\mathrm{Ba}} + B_{\mathrm{Kr}} - B_{\mathrm{U}}.$$

Hence

$$\boxed{B_{\mathrm{U}} = 1145.36 + 764.77 - 173.28 = 1736.85\,\mathrm{MeV}},$$

or 7.39 MeV per nucleon.

(iii)(a)

Problem

Calculate the energy released per kilogram of uranium-235 in the fission reaction above.

Solution

One fission releases

$$E_f = 173.28\,\mathrm{MeV} = 2.776 \times 10^{-11}\,\mathrm{J}.$$

Using $m_U \simeq 235\mathrm{u}$,

$$\boxed{\mathcal{E}_U = \frac{E_f}{235\mathrm{u}} = 7.11 \times 10^{13}\,\mathrm{J\,kg^{-1}}}$$

(iii)(b)

Problem

Natural gas has energy density $4.9 \times 10^{7}\,\mathrm{J\,kg^{-1}}$. By what factor is uranium-235 more energy-dense?

Solution

$$\boxed{\frac{\mathcal{E}_U}{\mathcal{E}_{\text{gas}}} = 1.45 \times 10^6}$$

(iii)(c)

Problem

If the UK's annual energy consumption is approximately 10^{19} J, find the equivalent ideal volume of uranium-235 fuel. Use uranium density $19.1\,\text{g}\,\text{cm}^{-3}$.

Solution

$$m = \frac{10^{19}}{7.11 \times 10^{13}} = 1.41 \times 10^5\,\text{kg},$$

$$\boxed{V = \frac{m}{\rho} = 7.36\,\text{m}^3}$$

This is a cube of side 1.95 m, ignoring efficiency, enrichment and incomplete burn-up.

(iv)

Problem

The D–T reaction

$$^2_1\text{H} + {}^3_1\text{H} \to {}^4_2\text{He} + {}^1_0\text{n}$$

releases 17.59 MeV. Find the tritium binding energy given $B(^4\text{He}) = 27.27\,\text{MeV}$ and deuterium mass 2.0141 u.

Solution

For deuterium,

$$B_D = (m_p + m_n - 2.0141\text{u})c^2 = 1.73\,\text{MeV}.$$

Since

$$17.59 = B_{\text{He}} - B_D - B_T,$$

$$\boxed{B_T = 27.27 - 1.73 - 17.59 = 7.95\,\text{MeV}}$$

(v)

Problem

Calculate the energy density in $\text{J}\,\text{kg}^{-1}$ of D–T fuel for the stated fusion reaction, using the combined deuterium and tritium mass. Hence find the ideal fuel mass in tonnes needed to supply 10^{19} J.

Solution

The reactant mass per event is $(2.0141 + 3.016)$u. Therefore

$$\boxed{\mathcal{E}_{DT} = \frac{17.59\,\text{MeV}}{5.0301\text{u}} = 3.37 \times 10^{14}\,\text{J}\,\text{kg}^{-1}}$$

For 10^{19} J,

$$\boxed{m = 2.96 \times 10^4\,\text{kg} = 29.6\,\text{tonnes}}$$

(vi)(a)

Problem

Assume D–T fusion becomes spontaneous when the Coulomb potential energy of two singly charged nuclei at separation 10^{-15} m equals their mean thermal kinetic energy. Estimate the required temperature using the constants supplied.

Solution

$$U = \frac{e^2}{4\pi\varepsilon_0 r} = 2.31 \times 10^{-13}\,\mathrm{J} \approx 1.44\,\mathrm{MeV}.$$

Equating $U = 3k_B T/2$ gives

$$\boxed{T = 1.11 \times 10^{10}\,\mathrm{K}}$$

(vi)(b)

Problem

Compare the classical ignition estimate with the Sun's core temperature, 1.5×10^7 K, and explain why fusion nevertheless powers the Sun.

Solution

The classical estimate is about 740 times larger. Solar fusion remains possible because nuclei quantum-tunnel through the Coulomb barrier; the Maxwell–Boltzmann high-energy tail, high density, enormous particle number and stellar timescale make rare successful encounters sufficient.

(vii)

Problem

Use the Semi-Empirical Mass Formula, with the coefficients supplied in the sheet, to predict the binding energy of ${}^{239}_{94}\mathrm{Pu}$.

Solution

We use

$$B = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} + \delta,$$

with $a_V = 15.56$, $a_S = 17.81$, $a_C = 0.711$, $a_A = 23.702$, $a_P = 34.0\,\mathrm{MeV}$. Since $A = 239$ is odd, $\delta = 0$. Evaluation gives

$$\boxed{B({}^{239}\mathrm{Pu}) = 1773.42\,\mathrm{MeV}}, \qquad \boxed{B/A = 7.420\,\mathrm{MeV}}$$

Computational link: This SEMF expression is well suited to a spreadsheet or vectorised parameter sweep: integer A, Z inputs generate B and B/A, with parity logic selecting the pairing term.

(viii)

Problem

For

$${}^{239}_{94}\mathrm{Pu} + {}^{1}_{0}\mathrm{n} \to {}^{134}_{54}\mathrm{Xe} + {}^{103}_{40}\mathrm{Zr} + 3\,{}^{1}_{0}\mathrm{n},$$

use $B/A = 8.4137\,\mathrm{MeV}$ for xenon-134 and the SEMF for zirconium-103. Predict the fission energy and the corresponding energy per kilogram of plutonium-239.

Solution

$$B_{\text{Xe}} = 134(8.4137) = 1127.44\,\text{MeV},$$

$$\boxed{B_{\text{Zr}} = 852.99\,\text{MeV}}$$

Thus

$$Q = B_{\text{Xe}} + B_{\text{Zr}} - B_{\text{Pu}} = \boxed{207.0\,\text{MeV}}$$

Per kilogram of plutonium-239,

$$\boxed{\mathcal{E} = 8.36 \times 10^{13}\,\text{J}\,\text{kg}^{-1}}$$

The occasional "plutonium-235" wording in this subpart is a typo; the stated reaction is plutonium-239.

(ix)(a)

Problem

For the aneutronic reaction

$$^2_1\text{H} + {}^6_3\text{Li} \to 2\,{}^4_2\text{He},$$

use the SEMF to predict the lithium-6 binding energy and hence the reaction energy. Compare with the measured 22.4 MeV.

Solution

For $^6_3$Li, both Z and N are odd, so $\delta = -a_P A^{-3/4}$. The SEMF gives

$$\boxed{B(^6\text{Li}) = 23.34\,\text{MeV}}$$

Therefore

$$Q_{\text{SEMF}} = 2(27.27) - 1.73 - 23.34 = \boxed{29.47\,\text{MeV}}$$

This exceeds the stated actual value 22.4 MeV by about 31.6%, illustrating the liquid-drop model's weakness for very light nuclei.

(ix)(b)

Problem

Explain why D–Li-6 fusion may be harder to achieve in a reactor than D–T fusion, despite avoiding energetic neutron emission.

Solution

The Coulomb barrier scales approximately with $Z_1 Z_2/r$. D–T has $Z_1 Z_2 = 1$, whereas D–Li has $Z_1 Z_2 = 3$. At a given temperature its tunnelling probability is therefore much lower, and lithium's larger mass also reduces thermal speed. Aneutronic products do not imply easy ignition.

(x)

Problem

Using the SEMF with $A = 2Z$, ignore pairing and find the integer Z that maximises B/A. Identify the corresponding element.

Solution

The asymmetry term vanishes. Substituting $A = 2Z$ into the SEMF and scanning positive integer Z gives

$$\boxed{Z = 25, \quad A = 50, \quad B/A \approx 8.410\,\text{MeV}}$$

Atomic number 25 is manganese: $\boxed{^{50}_{25}\text{Mn}}$
The imposed $N = Z$ constraint and omitted pairing term explain why this is not the empirical global maximum.

Question 2

Problem

The neutron-capture reaction

$$^{239}_{94}\text{Pu} + {}^{1}_{0}\text{n} \to {}^{240}_{94}\text{Pu} + \gamma$$

occurs approximately 27% of the time. Use the SEMF to calculate the change in binding energy and hence the smallest possible gamma-ray wavelength. Use $h = 6.63 \times 10^{-34}\,\text{J s}$.

Solution

The incident neutron is unbound, so

$$Q = B(^{240}\text{Pu}) - B(^{239}\text{Pu}).$$

For $^{240}_{94}\text{Pu}$, both Z and N are even, giving a positive pairing term. The SEMF gives

$$B_{240} = 1779.92\,\text{MeV}, \qquad Q = \boxed{6.50\,\text{MeV}}$$

If one photon carries all the released energy,

$$\boxed{\lambda_{\text{min}} = \frac{hc}{Q} = 1.91 \times 10^{-13}\,\text{m} = 0.191\,\text{pm}}$$

Recoil or multiple photons would increase the wavelength of each photon.

Computational link: The same SEMF calculator used for the binding-energy curves can evaluate neighbouring isotopes automatically, making Q-value differences and parity effects easy to verify.

Question 3

Problem

Construct a spreadsheet model of the SEMF and plot B/A against Z for $A = \text{round}(2.1Z)$, $A = 2Z$ and $A = \text{round}(1.9Z)$, over $1 \le Z \le 118$. Overlay and colour-code the three curves with a legend, retaining the pairing term.

Solution

The spreadsheet should contain Z, each rounded A, the parity-dependent pairing term, total B, and B/A. The three series are then overlaid on one scatter graph with connected lines.

The verified maxima are

A/Z	$Z_{\max}$	$A_{\max}$	$(B/A)_{\max}/\mathrm{MeV}$	element
2.1	25	53	8.600	Mn
2.0	24	48	8.448	Cr
1.9	24	46	8.205	Cr

The $A/Z = 2.1$ maximum is at $Z = 25$, not 26, when ordinary spreadsheet rounding is applied. Small odd–even oscillations come from the pairing term. At large Z, neutron-rich nuclei are favoured because extra neutrons add strong-force attraction without additional Coulomb repulsion.

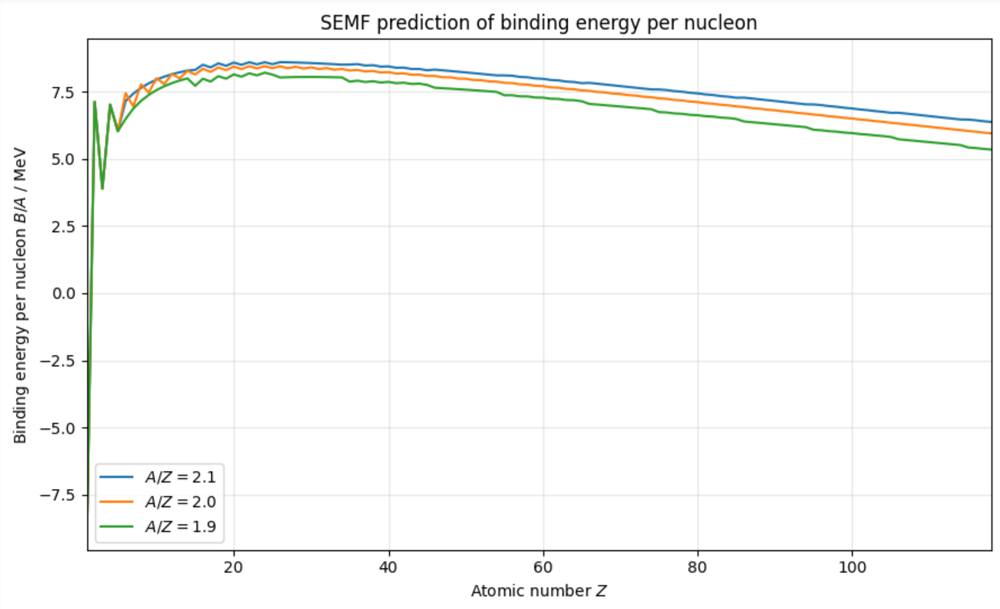

Figure 11: SEMF binding energy per nucleon for the three rounded mass-number rules.

The broad peak explains both fusion of light nuclei and fission of heavy nuclei: each process moves products towards larger binding energy per nucleon.

Computational link: This is a direct data-processing workflow: generate a parameter table, apply discrete rounding and parity logic, plot all series together, and use an automated maximum search to verify the optimum.

Question 4

Problem

Assume D–T fusion releases 17.59 MeV and, unrealistically, that all of it becomes kinetic energy of the fully ionised helium-4 nucleus, which is expelled directly backwards. Determine the relativistic mean exhaust speed and the D–T fuel-consumption rate required to produce 43.7 kN of thrust, equal to an Apollo AJ10 engine.

Solution

For $m_\alpha = 4.0026\mathrm{u} = 6.646 \times 10^{-27}\,\mathrm{kg}$,

$$K_\alpha = 17.59\,\mathrm{MeV} = 2.818 \times 10^{-12}\,\mathrm{J}.$$

Classically, $v = 2.91 \times 10^7\,\mathrm{m\,s^{-1}}$. Relativistically,

$$\gamma = 1 + \frac{K_\alpha}{m_\alpha c^2} = 1.00472,$$

$$\boxed{v = c\sqrt{1-\gamma^{-2}} = 2.90 \times 10^7\,\mathrm{m\,s^{-1}}}$$

The momentum per alpha particle is

$$p_\alpha = \gamma m_\alpha v = 1.94 \times 10^{-19}\,\mathrm{kg\,m\,s^{-1}}.$$

Thus the reaction rate is

$$\dot{N} = \frac{F}{p_\alpha} = 2.26 \times 10^{23}\,\mathrm{s^{-1}}.$$

Each reaction consumes $(2.0141 + 3.016)\mathrm{u} = 8.35 \times 10^{-27}\,\mathrm{kg}$, so

$$\boxed{\dot{m}_{DT} = 1.89 \times 10^{-3}\,\mathrm{kg\,s^{-1}} = 1.89\,\mathrm{g\,s^{-1}}}$$

This corresponds to about $6.8\,\mathrm{kg\,h^{-1}}$. The assumptions are extremely optimistic: in real D–T fusion most energy goes to the neutron, which cannot be magnetically redirected, and confinement, conversion and collimation are imperfect.

Computational link: The thrust estimate can be implemented as a parameter model in which fusion energy, particle mass and relativistic momentum are varied; the mass-flow output provides an engineering extension.

Radioactivity

Question 1

(i)

Problem

A radon sample falls to 2% of its original amount in 22 days. Determine its half-life in days.

Solution

$$0.02 = \mathrm{e}^{-22\lambda} \quad \Rightarrow \quad \lambda = -\frac{\ln 0.02}{22} = 0.1778\,\mathrm{d^{-1}}.$$

Thus

$$\boxed{t_{1/2} = \frac{\ln 2}{\lambda} = 3.90\,\mathrm{d}}$$

Computational link: A deterministic exponential curve can be compared with a Monte Carlo simulation of individual random decays, reusing the random-sampling approach of Task 1.

(ii)

Problem

Carbon-14 has half-life 5370 years. Fresh biomatter gives $238\,\mathrm{Bq\,kg^{-1}}$, while a reindeer-horn hammer gives $1.05\,\mathrm{Bq\,kg^{-1}}$. Estimate the hammer's age.

Solution

Activity follows the same law as population:

$$A = A_0 2^{-t/t_{1/2}}.$$

Therefore

$$\boxed{t = 5370 \log_2 \left(\frac{238}{1.05} \right) = 4.20 \times 10^4 \,\text{yr}}$$

The result assumes unchanged initial activity, a closed sample and no contamination.

(iii)

Problem

A 200 Bq strontium-90 beta source is measured through 2.00 mm of aluminium, reducing the measured rate to 10 Bq. Calculate the aluminium half-thickness and attenuation constant μ.

Solution

With $I = I_0 \mathrm{e}^{-\mu x}$,

$$\mu = \frac{1}{x} \ln \frac{I_0}{I} = \boxed{1.50\,\text{mm}^{-1} = 1.50 \times 10^3\,\text{m}^{-1}}$$

The half-thickness is

$$\boxed{x_{1/2} = \frac{\ln 2}{\mu} = 0.463\,\text{mm}}$$

Computational link: The same exponential-model code can use material thickness as the independent variable; a log-linear plot returns $-\mu$ as the gradient.

(iv)

Problem

Radium-226 has half-life 1602 years. Find the mass in a 9900 Bq source. Find the mass of uranium-235 with the same activity if its half-life is 2.22×10^{16} s.

Solution

Using $A = \lambda N$, $\lambda = \ln 2/t_{1/2}$ and $m = NM/N_A$,

$$\boxed{m_{\text{Ra}} = 2.71 \times 10^{-10}\,\text{kg} = 0.271\,\mu\text{g}},$$

$$\boxed{m_{\text{U}} = 1.24 \times 10^{-4}\,\text{kg} = 0.124\,\text{g}}$$

The much longer U-235 half-life gives a much lower specific activity.

(v)

Problem

Protactinium-231 decays to actinium-227, and then through short-lived intermediates to stable lead-207. If a pure Pa-231 object is eventually 90% lead and $t_{1/2} = 32\,800$ years, estimate the elapsed time.

Solution

Treating the short-lived daughters as effectively immediate, 10% Pa remains:

$$0.10 = 2^{-t/32800} \quad \Rightarrow \quad \boxed{t = 32800 \log_2 10 = 1.09 \times 10^5\,\text{yr}}$$

Question 2

(i)

Problem

Iodine-123 has half-life 13.22 hours and decays by electron capture to very stable tellurium-123, releasing a 159 keV gamma ray. Given $Z_{\text{I}} = 53$, write the nuclear equation.

Solution

$$\boxed{{}^{123}_{53}\text{I} + {}^{0}_{-1}e \to {}^{123}_{52}\text{Te} + \nu_e + \gamma}$$

More explicitly, the capture can produce an excited daughter, followed by the 159 keV gamma de-excitation.

(ii)

Problem

State the properties of iodine-123 that make it useful for medical applications.

Solution

Its 159 keV gamma rays escape tissue and suit gamma-camera detection; its 13.22 h half-life is long enough for preparation and imaging but short enough to limit prolonged activity; electron capture avoids a strong beta-particle dose; iodine is biologically concentrated in the thyroid; and the daughter is very stable.

(iii)

Problem

An initial activity 2.5×10^7 Bq is administered for thyroid imaging. Calculate the total activity after 24 hours and explain why the activity in the thyroid itself is smaller.

Solution

$$A = A_0 2^{-24/13.22} = \boxed{7.10 \times 10^6\,\text{Bq} = 7.10\,\text{MBq}}$$

Thyroid activity is lower because uptake is incomplete and biological clearance removes iodine from the gland and body.

(iv)

Problem

Calculate the number of iodine-123 atoms in the initial dose.

Solution

$$\lambda = \frac{\ln 2}{13.22(3600)} = 1.457 \times 10^{-5}\,\text{s}^{-1},$$

$$\boxed{N_0 = \frac{A_0}{\lambda} = 1.72 \times 10^{12}}$$

(v)

Problem

Calculate the total energy in joules released in gamma emissions if all iodine-123 nuclei decay. State the assumption required to interpret this as absorbed energy.

Solution

$$E_\gamma = 159\,\text{keV} = 2.547 \times 10^{-14}\,\text{J},$$

$$\boxed{E_{\text{total}} = N_0 E_\gamma = 4.37 \times 10^{-2}\,\text{J}}$$

This is the emitted energy and therefore an upper bound on absorbed energy; some gamma photons leave the patient.

(vi)

Problem

Determine the gamma-emission power immediately after administration and compare it with the average power over 48 hours.

Solution

$$\boxed{P_0 = A_0 E_\gamma = 6.37 \times 10^{-7}\,\text{W} = 0.637\,\mu\text{W}}$$

The energy emitted by time T is $N_0[1 - 2^{-T/t_{1/2}}]E_\gamma$. Hence

$$\boxed{\bar{P}_{48} = 2.33 \times 10^{-7}\,\text{W} = 0.233\,\mu\text{W}},$$

about 36.5% of the initial power.

Computational link: A time-stepped decay model can plot both instantaneous power $P(t)$ and accumulated emitted energy, avoiding confusion between initial and average power.

Question 3

(i)

Problem

Americium-241 decays by alpha emission to neptunium. Given $A = 241$ and $Z = 95$, write the nuclear equation.

Solution

$$\boxed{{}^{241}_{95}\text{Am} \to {}^{237}_{93}\text{Np} + {}^{4}_{2}\text{He}}$$

(ii)

Problem

A smoke detector contains $0.33\,\mu$g of AmO_2. Using $t_{1/2} = 432.2$ years and oxygen mass number 16, calculate the americium activity in becquerels and curies.

Solution

The molar mass is approximately $273\,\mathrm{g\,mol^{-1}}$, so

$$N = \frac{0.33 \times 10^{-6}}{273} N_A = 7.28 \times 10^{14}.$$

With $\lambda = 5.08 \times 10^{-11}\,\mathrm{s^{-1}}$,

$$\boxed{A = 3.70 \times 10^4\,\mathrm{Bq} = 37.0\,\mathrm{kBq}}$$

Since $1\,\mathrm{Ci} = 3.7 \times 10^{10}\,\mathrm{Bq}$, this is

$$\boxed{1.00\,\mu\mathrm{Ci}}$$

(iii)

Problem

Each alpha particle has energy $5.486\,\mathrm{MeV}$, and the mean ionisation energy of air is $34\,\mathrm{eV}$. Estimate the ionisations per second and hence the ideal smoke-free detector current.

Solution

One $5.486\,\mathrm{MeV}$ alpha can ideally produce

$$\frac{5.486 \times 10^6}{34} = 1.61 \times 10^5$$

ion pairs. Therefore

$$\boxed{\dot{N}_{\mathrm{ion}} = 5.97 \times 10^9\,\mathrm{s^{-1}}}$$

Collecting one elementary charge per ion pair gives

$$\boxed{I = e\dot{N}_{\mathrm{ion}} = 9.56 \times 10^{-10}\,\mathrm{A} = 0.956\,\mathrm{nA}}$$

This is an upper estimate because not all alpha energy creates collectable ion pairs and recombination occurs.

(iv)

Problem

The current falls to 20% of its smoke-free value when charge attaches to smoke particles. Taking dry-air molar mass as $28.97\,\mathrm{g\,mol^{-1}}$ and assuming equal temperature, estimate an effective smoke-particle molar mass.

Solution

Under the worksheet's simplified model, $I \propto v$ and equal-temperature thermal speeds satisfy $v \propto M^{-1/2}$. Thus

$$0.20 = \sqrt{\frac{28.97}{M_{\mathrm{smoke}}}},$$

so

$$\boxed{M_{\mathrm{smoke}} = 724\,\mathrm{g\,mol^{-1}}}$$

This is only an effective estimate: real charge mobility depends strongly on particle size, shape, drag, collisions and electric field, not mass alone.

Question 4

(i)

Problem

Near the Chernobyl Elephant's Foot, the dose rate was $80\,\mathrm{Gy\,h^{-1}}$. Assuming typical alpha or beta energy $5\,\mathrm{MeV}$, calculate the number absorbed per second by a $10\,\mathrm{cm}$ cube of water. How does the flux change if the cube is moved three times farther away?

Solution

The cube has volume $10^{-3}\,\mathrm{m^3}$ and mass $1\,\mathrm{kg}$. Hence its absorbed power is

$$P = \frac{80\,\mathrm{J}}{3600\,\mathrm{s}} = 2.22 \times 10^{-2}\,\mathrm{W}.$$

For $E_p = 5\,\mathrm{MeV} = 8.01 \times 10^{-13}\,\mathrm{J}$,

$$\boxed{\dot{N} = 2.77 \times 10^{10}\,\mathrm{s^{-1}}}$$

Using the $0.01\,\mathrm{m^2}$ exposed face,

$$\boxed{\Phi = 2.77 \times 10^{12}\,\mathrm{m^{-2}\,s^{-1}}}$$

For an approximately localised isotropic source, $\Phi \propto r^{-2}$, so at $3r$,

$$\boxed{\Phi' = \Phi/9 = 3.08 \times 10^{11}\,\mathrm{m^{-2}\,s^{-1}}}$$

The calculation follows the worksheet's idealised particle-energy model; the actual external field is dominated by penetrating radiation, not alpha particles travelling macroscopic distances through air.

(ii)

Problem

Model the Elephant's Foot as a spherical cap of base radius $a = 1.0\,\mathrm{m}$ and height $h = 0.5\,\mathrm{m}$. Assuming its surface flux equals that calculated previously, estimate its total activity. Also prove using calculus that the curved surface area is

$$A = \pi(a^2 + h^2).$$

Solution

The curved area is

$$A = \pi(a^2 + h^2) = 1.25\pi = 3.93\,\mathrm{m^2}.$$

Assuming the surface flux equals that above,

$$\boxed{\mathcal{A} = \Phi A = 1.09 \times 10^{13}\,\mathrm{Bq}}$$

This assumes one relevant particle per decay, uniform surface flux, complete energy absorption and unchanged geometry.

For the calculus proof, rotate $y = \sqrt{R^2 - x^2}$ about the x-axis from $x = R - h$ to R:

$$A = 2\pi \int_{R-h}^{R} y\sqrt{1 + (y')^2}\, dx.$$

Since $y' = -x/\sqrt{R^2 - x^2}$, the integrand simplifies to R, giving

$$A = 2\pi Rh.$$

At the cap base, $a^2 + (R - h)^2 = R^2$, so $a^2 + h^2 = 2Rh$. Therefore

$$\boxed{A = \pi(a^2 + h^2)}$$

(iii)

Problem

A 4.2 mm thickness of lead halves the gamma count rate. Find the lead thickness needed to reduce an unshielded 2.77×10^8 Bq count to approximately 5 Bq.

Solution

$$n = \log_2\left(\frac{2.77 \times 10^8}{5}\right) = 25.72$$

half-value layers, so

$$\boxed{x = 25.72(4.2\,\mathrm{mm}) = 108\,\mathrm{mm} \approx 10.8\,\mathrm{cm}}$$

Rounding up to 26 complete layers gives 109.2 mm.

Computational link: A shielding calculator can loop over integer half-value layers and report the first thickness that satisfies a target dose rate, rather than relying only on continuous rounding.

Question 5

(i)

Problem

An RTG contains 4.8 kg of plutonium-238, with half-life 87.7 years. Each alpha has energy 5.593 MeV. Calculate the number of atoms, initial activity and ideal thermal power, assuming all alpha kinetic energy becomes heat.

Solution

$$N_0 = \frac{4.8}{0.238} N_A = \boxed{1.21 \times 10^{25}}$$

With $t_{1/2} = 87.7\,\mathrm{yr} = 2.767 \times 10^9\,\mathrm{s}$,

$$\lambda = 2.505 \times 10^{-10}\,\mathrm{s}^{-1}, \qquad \boxed{A_0 = 3.04 \times 10^{15}\,\mathrm{Bq}}$$

Each 5.593 MeV alpha carries 8.96×10^{-13} J, so

$$\boxed{P_0 = A_0 E_\alpha = 2.72 \times 10^3\,\mathrm{W}}$$

The ideal specific thermal power is $\boxed{568\,\mathrm{W\,kg^{-1}}}$
Electrical output is lower because thermoelectric conversion is inefficient.

(ii)

Problem

Calculate and plot the thermal power per kilogram of plutonium-238 against time in years.

Solution

$$\boxed{\frac{P(t)}{m} = 568\,2^{-t/87.7}\,\mathrm{W\,kg^{-1}}},$$

with t in years. Representative values are

t/yr	0	87.7	175.4	263.1	350.8	438.5
P/m	568	284	142	71.0	35.5	17.8

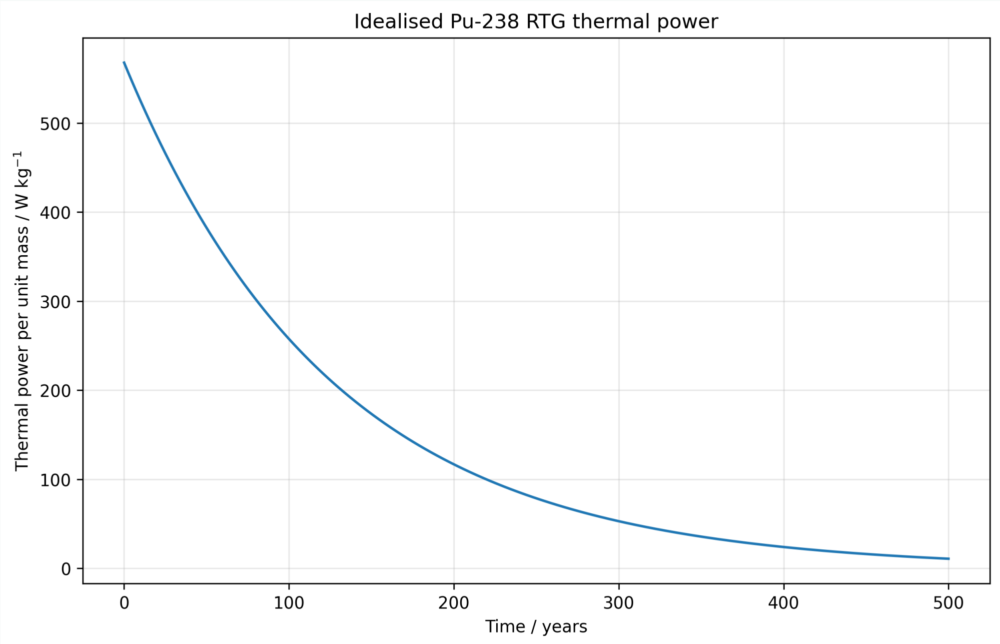

Figure 12: Idealised Pu-238 thermal power per kilogram.

Computational link: This is naturally presented as a time-series model with half-life, initial mass and specific energy as adjustable parameters.

Question 6

(i)

Problem

Construct a spreadsheet to verify the Geiger–Nuttall law using

Isotope	Z	$t_{1/2}/\mathrm{s}$	E_α/MeV
Po-212	84	2.99×10^{-7}	8.955
Po-214	84	1.60×10^{-4}	7.680
Rn-218	86	3.50×10^{-2}	7.263
At-217	85	3.20×10^{-2}	7.000
Bi-213	83	2.74×10^{3}	5.870
Ra-224	88	3.14×10^{5}	5.789
Th-238	90	6.03×10^{7}	5.520
Pa-231	91	1.03×10^{12}	5.150
Np-237	93	6.75×10^{13}	4.959
Th-232	90	4.43×10^{17}	4.081

Plot

$$y = \log_{10}(t_{1/2}/\mathrm{s}) \quad \text{against} \quad x = \frac{Z}{\sqrt{E_\alpha/\mathrm{MeV}}},$$

fit a straight line, and compare its gradient and intercept with the stated law.

Solution

Create transformed columns

$$x = \frac{Z}{\sqrt{E_\alpha/\mathrm{MeV}}}, \qquad y = \log_{10}(t_{1/2}/\mathrm{s}),$$

and fit $y = mx + c$. Linear regression gives

$$\boxed{m = 1.5077, \qquad c = -49.277},$$

so

$$\boxed{\log_{10}\left(\frac{t_{1/2}}{\mathrm{s}}\right) \approx 1.508\frac{Z}{\sqrt{E_\alpha/\mathrm{MeV}}} - 49.277}$$

The logarithm is essential because the half-lives span roughly 24 orders of magnitude.

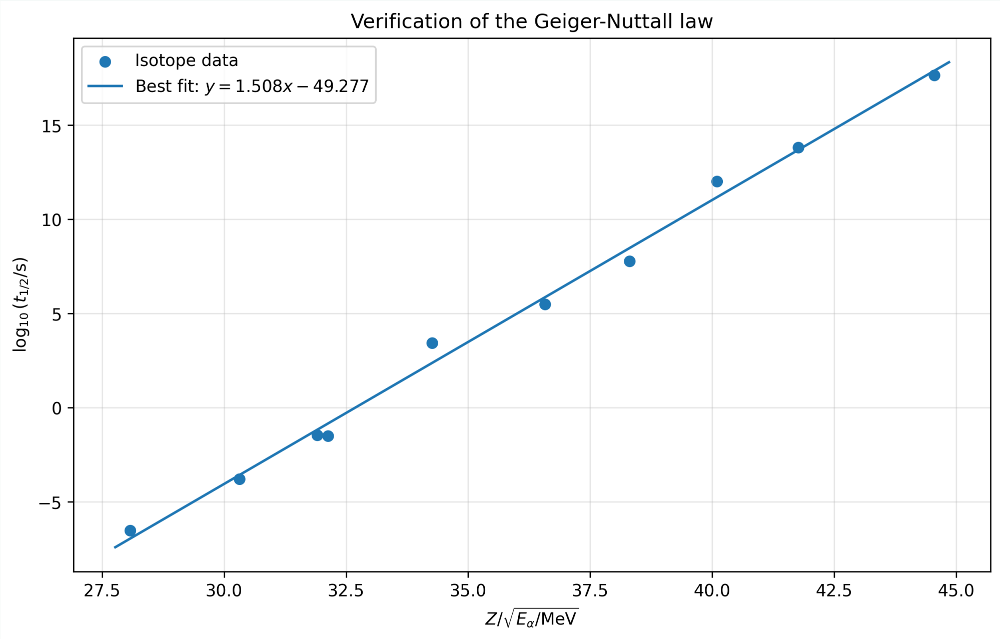

Figure 13: Linearised Geiger–Nuttall data and fitted line.

Computational link: This follows the challenge's data-analysis workflow: transform the raw variables, fit a straight line, inspect residuals and use the fitted parameters for prediction.

(ii)

Problem

Verify that

$$\log_{10}\left(\frac{t_{1/2}}{\mathrm{s}}\right) \approx \frac{1.51Z}{\sqrt{E_\alpha/\mathrm{MeV}}} - 49.3,$$

and use it to predict the half-life of francium-221, where $Z = 87$ and $E_\alpha = 6.3\,\mathrm{MeV}$.

Solution

$$\log_{10}(t_{1/2}/\mathrm{s}) = 3.039,$$

so

$$\boxed{t_{1/2} = 1.09 \times 10^3\,\mathrm{s} = 18.2\,\mathrm{min}}$$

The law is empirical; nuclear structure and angular momentum produce isotope-specific deviations.

Computational link: The fitted line can be wrapped into a predictor that accepts Z and E_α, while uncertainty bands show the limits of the empirical law.